The usual statistical analysis performed on experimental data measured in these devices makes use of the Weibull distribution (WD) [1, 19, 20]; nevertheless, sometimes its fit with experimental data is not very accurate. In the next section, it will be shown that WD does not work correctly for the devices under consideration in this manuscript. In fact, recent dielectric breakdown studies have shown that the WD does not describe the stochastic trends well enough, more so in downscaled structures at the low and high percentile regions. The validity of a defect clustering model for RRAM switching parameters was recently examined [21].

Therefore, another statistical approach is needed. Apart from an accurate statistical description of experimental data, the interpretation of the parameters extracted by the application of the statistical analysis can shed new light to the variability issue and the physics behind RRAM operation.

In order to deepen on this issue, a thorough analysis of the statistical properties of RRAM variability is performed. To do so, phase-type distributions (PHDs) will help us to analyse the possible intermediate states of degradation in the conductive filament destruction processes that lead to a RRAM high resistivity state (the rupture of the conductive filament isolates the electrodes and therefore the RRAM resistance increases).

Phase-type distributions, which were introduced and analyzed in detail by Neuts [22, 23, 24], constitute a class of non-negative distributions that makes it possible to model complex problems with well-structured results, thanks to its matrix-algebraic form. Due to their valuable properties, many varieties of this class of distributions have been considered in diverse branches of science and engineering and applied in reliability studies. Particular cases of PHDs are the exponential, Erlang, generalized Erlang, hypergeometric and Coxian distributions, among others. In fact, not only very well-known probability distributions are PHDs but also any nonnegative probability distribution can be approximated as needed taking into consideration that the PHD class is dense in the set of probability distributions on the nonnegative half-line [25]. The more essential and important properties of PHDs were reviewed in a recent study [26].

As reported below, the versatility and advantages of the PHD will come up to allow a better analysis of RRAM variability. In this context, it will be shown that the Erlang distribution (ED) (a particular PHD) works much better than WD to describe the experimental data under consideration in this manuscript. The physical interpretation of the fitted parameters from the PHD modelling will shed light on the explanation of RRAM variability.

The fabricated devices and measurement process are described in Section 2, the new statistical approach for modelling RRAMs variability is given in Section 3 and the main results and discussion in Section 4. Finally, the conclusions are given in Section 5.

2. DEVICE DESCRIPTION AND MEASUREMENT

The devices employed in this manuscript are unipolar Ni/HfO2/Si-based RRAMs. The fabrication details were given in [27]. A HP-4155B semiconductor parameter analyser was used in the measurement process that consisted of a long series of RS cycles under ramped voltage stress. The Si substrate (bottom electrode) was grounded and a negative voltage was applied to the Ni (top electrode), although for simplicity we have assumed the absolute value of the applied voltage henceforth [27].

Three reset current versus voltage curves of the series of resistive switching cycles (2749 cycles) measured are shown in Figure 1. In the curves plotted, a sudden current drop can be seen corresponding with the rupture of the conductive filament (highlighted as reset point [1, 5, 27, 14, 15]) that connects the electrodes (in this respect, the conductive filament works as a fuse). The corresponding voltages and currents are known as reset voltages and currents respectively [1, 18], they have been explicitly shown in Figure 1 for the sake of clarity. The reset voltage determination has been performed by detecting a 50% current variation at the reset point (when the sudden current drop takes place). Others methods have been proposed in the literature [18]; nevertheless, this one worked well for the devices under consideration here.

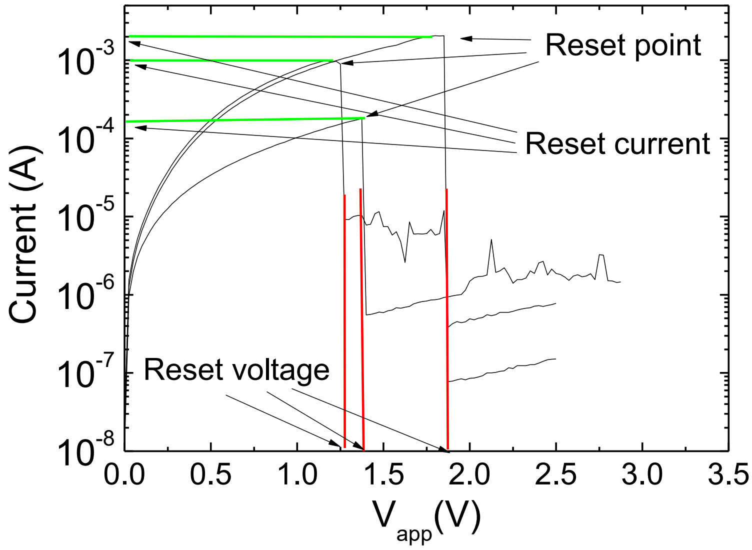

Figure 1. Experimental current versus applied voltage (shown in black lines) for three curves of a 2749 series of continuous resistive switching cycles, including set and reset curves. The reset point and the corresponding reset voltages (V_{reset}, indicated by vertical red lines) and reset currents (I_{reset}, indicated by horizontal green lines) are shown for clearness.

As noted in the introduction, the V_{reset} and I_{reset} distributions (as well as V_{set} and I_{set} distributions) have been subject of a deep statistical analysis for many different RRAMs [20, 28]. The WD was employed to describe the statistical properties of experimental data when the reset and set parameters were extracted. The Weibull model has been successfully employed along with a geometrical cell-based model which was connected with the percolation model for oxide breakdown for SiO_2-based devices [29]. In